

RE-DESIGNING OUR DAY-TO-DAY APP

The team-based, semester-long project is to think deeply on design-related concepts, principles and guidelines and apply them in a specific context, as well as practice the techniques for usability evaluation to identify the design bugs and refine the interaction accordingly.

DESCRIPTION

The overall project is a semester-long one, and can be roughly titled as “Re-designing our day-to-day interaction by applying [a] to support [b].” The project is to take a user-interface of a commonly-used app, web service or any other interactive system, and significantly revise the way the user is to interact with it by applying one particular design concept/principle/guideline for a particular purpose in mind. The result will be not a bland, usual or regular UI but somewhat unusual UI not commonly seen due to the emphasis on a particular angle of design optimized for a specific situation/user/context.

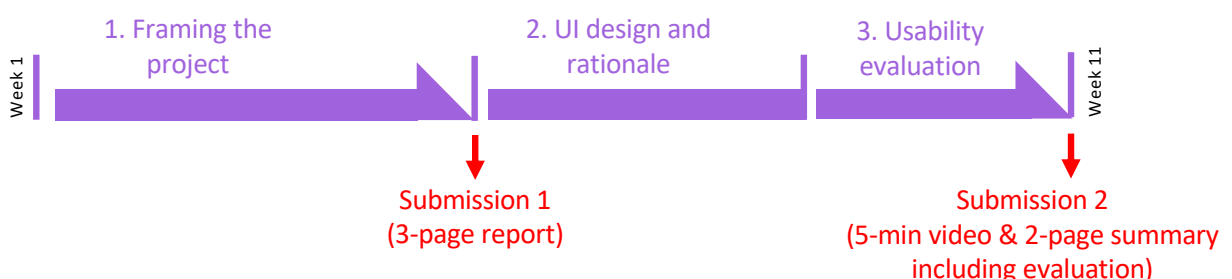


Diagram above shows the 3 components of the projects and 2 submissions. Both submissions in the project are group submissions, not individual submission.

PROJECT

TEAM FORMATION

Students will be self-forming the groups as a team of 5-6 members by Week 3 (check exact deadline on Loop), and those who could not form will be randomly grouped by the lecturer. The team members will receive the same grades for all their submissions (unless there is an exceptional situation), as it is expected that all members will equally contribute to the successful completion of the project.

TIPS & SUGGESTIONS

Since each lecture will be full of exciting contents on how to design a good user-interface (and what on earth that means) and how we can say it is well designed, there are many ways the project can be shaped, framed and executed. Avoid procrastination and try to push the tasks as early as possible. Don't be late on the submission as there will be heavy penalty for late submission. Keep sketching the UI and keep refining – each sketch is an iteration in design and being able to see it so that it can be further refined is a very important aspect of design. Keep the interim design sketches, so that they can be revisited, documented and submitted as appendix as an evidence of progress.

The two group submissions contribute to total of 35% of the module grade, divided into 2 temporally-spaced submissions.

SUBMISSION 1: FRAMING THE PROJECT

Through 2 submissions, the team-based project will help apply the design concepts, principles and guidelines learned in a specific context. First one is to shape the project by deciding a particular concept/principle/guideline and to plan.

DESCRIPTION

During the first 4 weeks of the module, we will be introduced the key concepts related to user-interface design, usability principles and design guidelines. While we learn these, we can think about which one(s) will be a good topic to shape the design project. Once the team is formed, the members should regularly meet to discuss and decide which concept/principle/guideline to use to shape the project. This then becomes the starting point of the remaining project, and the first of the 2 group submissions. See what kind of topics are introduced in the first 3-4 weeks while tentatively select some of them, brainstorm how these could be the anchor to develop the project. Come up with a provisional title that represents the project idea, and also think about some rough idea of what the eventual user-interface might look like. This constitutes 15% of the module grade.

EXAMPLES

“Increased visibility to maximise the learnability of Photoshop for the beginners”: Photoshop is a popular image manipulation software, but due to so many functionalities represented on its UI it can be daunting to the first-time users. This project takes a design principle of “visibility” to completely hide certain aspects of functions and reveal other aspects

in such a way that will streamline how such a large amount of image-related functions are presented to the beginner users so that the efforts to learn how to use this powerful software in the first 2-3 months will be minimized.

“Personalised online purchasing experience to minimize tap distance on a smartphone” => measuring the click/tap distance is one of the common ways to assess a user-interface. This project will take a personalised information retrieval strategy to guess what the user wants/needs at a particular time/day in a particular location in a particular season, so that the user does not even need to search and select a product, thus significantly reducing the click/tap distance.

“Augmenting YouTube UI with within-video level bookmarking, commenting and sharing: analytic approach to optimise the within-video selection and follow-up actions” => task analysis and sequencing can be applied to add a novel feature of sub-video level operations that a user can do. Temporal data visualisation techniques will be systematically reviewed, classified to come up with a more sophisticated and information-rich timeline on the playback panel.

SUBMISSION

3-page report (excluding title page, table of contents, references, appendix, etc.) in PDF format.

Deadline: 11:59pm, 11th October 2026

TIPS & SUGGESTIONS

It will be good if the report **has a title that represents the chosen topic**. It will also be good to have some preliminary sketches or rough ideas drawn to suggest where the project will be going. Mention what area(s) will be the focus of the project, what might be key literature/book/paper to read. Mention how it will be prototyped, in what platform, in what level of fidelity. Mention how it will be evaluated.

PROJECT

There is room for change of direction after the submission, i.e. after the report was submitted and the lecturer grades it with any feedback, the team can take it into account to modify/shift/augment the topic accordingly.

GRADING RUBRICS

Use the rubrics below to strengthen the submission

Criteria	Marks
Title represents the project topic well	0 - 2
Clear and reasonable articulation of the chosen UI concept/principle/guideline	0 - 5
Arguments informed by facts/ literature/ survey/ references	0 - 2
Novelty	0 - 2
Solid plan for the project (platform, member contribution, etc.)	0 - 2
Effective communication of the idea (writing, presentation)	0 - 2

MEMBER CONTRIBUTION

Submission must include an extra page with a table of member contribution, to be explained during class. See example table below.

SUBMISSION 2: UI DESIGN RATIONALE & EVALUATION

Second part is the actual design and implementation of the UI, its design rationale, and the evaluation report

DESCRIPTION

The core component of the project is the delivery of a prototype UI designed based on the UI concept/principle/guideline. Major design decisions should be made largely based on the chosen key topic, while all other minor decisions should be based on what was learned throughout the classes. The UI should be the result of an **iterative sketching exercise and refinements**, and the final prototype needs to be **interactive** enough so that it can be realistically tested with users. It is important to note that the UI needs to clearly demonstrate that a particular concept/ principle/ guideline has been the guiding factor in its major design points.

How can you say your UI is doing what you designed it for? How can you say you succeeded in your design? Usability evaluation is the answer: the designed interactive UI will need to be assessed by employing the evaluation techniques learned in the latter part of the module. The evaluation activity should start by deciding what technique(s) to choose given the specific context the UI is set, the depth/breadth of the prototype, and what aspects need to be assessed. Taking User-Centred Design approach, some type of techniques that involve real users need to be used – recruit the suitable participants and conduct the testing with them while observing their usage, mistakes they are making, confusions they have, questions they have, etc. and also asking questions in a debriefing time. Report the whole process employed. We will learn the details of evaluation techniques in the last 4 weeks of the module. This submission constitutes 20% of the module grade.

EXAMPLE EVALUATION

The team developed an interactive banking app for smartphone designed to help the older adults more independent in banking. 5 older adults ages between 60 and 75 were recruited one by one, and each participant was given 3 specific tasks: (1) check the balance, (2) pay a bill, (3) transfer €20 from one account to another. The team member observed the participant perform each task and noted down anything worth noting. A short post-task questionnaire was administered to capture subjective satisfaction and other thoughts. These findings were then qualitative analysed to identify themes and issues, then these informed how the UI could/should be revised and refined.

SUBMISSION

Following must be submitted:

1. Video (maximum duration of 5 minutes) should be uploaded on a sharable video service (e.g. YouTube) and the link indicated in the written report (see below) for assessment. The video should demonstrate the concept(s) applied and the final design, interactivity, as well as the evaluation and its results.
2. Written report of 2 pages (excluding title page, references, appendix, etc.) summarising the designed UI, process, number of iterations, insights, and the evaluation methodology, analysis and findings/insights.
3. Member contribution: there should be an extra 1 page summarising individual contribution of each member (see example below)

Deadline: 11:59pm, 22nd November 2026

GRADING RUBRICS

Criteria	Marks
One chosen UI concept/principle/guideline impacting the shaping the UI	0 - 5
Adherence to other UI concepts/ principles/ guidelines throughout the UI	0 - 5
Choice and rationale for evaluation technique(s)	0 - 3
Efforts to ensure ecological validity	0 - 2
Interactivity/implementation	0 - 3
Presentation (video + report)	0 - 2

Example member contribution table

Member name	Overall contribution	Area of contribution
Pat Murphy	20%	Ideation, framing and concrete visual design, making video script/planning
Jane O'Connor	20%	Archiving the evolution of sketches, reflecting and generating insights; user testing planning
John Smith	20%	UI design and prototyping in Origami; making widget assets from Paint app; recruiting the participants
Orla Duffy	20%	Documenting the design rationale, references, working with Jane on evaluation planning
Kim Dunne	20%	UI concept championing, helping with framing (with Pat), proof-reading the report, lead for video making